OPTICAL MUSIC RECOGNITION FOR JAZZ LEAD SHEETS

A CRNN–CTC Approach for Monophonic Lines

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Resum

El *Reconeixement Òptic de Música* (OMR) és el procés d'obtenir representacions simbòliques llegibles per màquina a partir d'imatges de partitures. Malgrat els avenços en el camp, la transcripció automàtica de partitures reals de baixa qualitat continua essent un repte obert. Aquesta memòria presenta un sistema OMR orientat a línies de melodia monofòniques de jazz del recull *The Real Book*.

El sistema es basa en una xarxa neuronal recurrent convolucional amb pèrdua CTC (*CRNN-CTC*), entrenada sobre dades sintètiques generades a partir del conjunt PrIMuS. Per reduir la diferència entre imatges sintètiques i escaneigs reals, s'aplica una etapa d'augmentació que simula artefactes fotogràfics habituals. El model s'avalua amb la *Taxa d'Error de Símbols* (SER), assolint un **1,17%** al conjunt de prova sintètic, i un ajust fi sobre pàgines reals del Real Book millora el rendiment sobre imatges fora de domini.

El sistema complet integra detecció de pentagrames, reconeixement de melodia, extracció de símbols d'acords i correcció gramatical, i accepta fitxers PDF o imatges com a entrada. Aquest treball demostra la viabilitat dels models CRNN-CTC per al domini específic de les línies de jazz monofòniques i estableix les bases per a extensions futures cap a la transcripció polifònica.

Resumen

El *Reconocimiento Óptico de Música* (OMR) es el proceso de obtener representaciones simbólicas legibles por máquina a partir de imágenes de partituras. A pesar de los avances en el campo, la transcripción automática de partituras reales de baja calidad sigue siendo un desafío abierto. Esta memoria presenta un sistema OMR orientado a líneas de melodía monofónicas de jazz de la colección *The Real Book*.

El sistema se basa en una red neuronal recurrente convolucional con pérdida CTC (*CRNN-CTC*), entrenada sobre datos sintéticos generados a partir del conjunto PrIMuS. Para reducir la brecha entre imágenes sintéticas y escáneres reales, se aplica una etapa de aumento que simula artefactos fotográficos habituales. El modelo se evalúa con la *Tasa de Error de Símbolos* (SER), alcanzando un **1,17%** en el conjunto de prueba sintético, y un ajuste fino sobre páginas reales del Real Book mejora el rendimiento sobre imágenes fuera de dominio.

El sistema completo integra detección de pentagramas, reconocimiento de melodía, extracción de símbolos de acorde y corrección gramatical, y acepta archivos PDF o imágenes como entrada. Este trabajo demuestra la viabilidad de los modelos CRNN-CTC para el dominio específico de las líneas de jazz monofónicas y sienta las bases para extensiones futuras hacia la transcripción polifónica.

Abstract

Optical Music Recognition (OMR) is the task of automatically converting images of musical scores into machine-readable symbolic representations. Despite advances in the field, transcribing real-world scans of informal or low-quality sources remains an open challenge. This thesis presents an end-to-end OMR system targeting monophonic melody lines in jazz lead sheets, with a focus on the *Real Book* collection.

The system is built around a *Convolutional Recurrent Neural Network with Connectionist Temporal Classification* loss (CRNN-CTC), trained on synthetic data derived from the PrIMuS dataset. To bridge the gap between clean synthetic renderings and real scans, an augmentation stage simulates common photographic artefacts. Evaluated using the *Symbol Error Rate* (SER), the model achieves **1.17%** on the synthetic test set, and a fine-tuning stage on real Real Book pages further improves performance on out-of-domain images.

The full inference pipeline integrates staff detection, melody recognition, chord symbol extraction, and a grammar correction stage, accepting PDF or image input. This work demonstrates the feasibility of CRNN-CTC-based OMR for the specific domain of monophonic jazz lead sheets and lays the groundwork for future extensions towards polyphonic transcription.

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Chapter 1

Introduction and Context

Music notation is one of the most sophisticated symbolic languages ever devised: a two-dimensional graphical system capable of encoding pitch, rhythm, dynamics, phrasing, and expressive intent with sufficient precision to allow performers separated by centuries to reproduce a composer's intentions from a printed page. Yet the overwhelming majority of notated music exists today only as paper or low-resolution scanned image, inaccessible to the computational tools that now drive music education, music information retrieval (MIR), computational musicology, and digital archiving. *Optical Music Recognition* (OMR) is the discipline that bridges this divide: it translates images of musical scores into structured, machine-readable representations in the same way that Optical Character Recognition (OCR) transformed printed text into searchable, editable data [1].

This thesis describes the design, implementation, and quantitative evaluation of an end-to-end OMR system tailored to a concrete and practically significant target: the monophonic melody lines and chord symbols of jazz lead sheets, with a primary focus on *The Real Book*, the iconic fake-book in the hands of most working jazz musicians worldwide.

1.1 Motivation

1.1.1 The Digitization Gap in Music

Music archives, educational institutions, and individual musicians hold enormous quantities of repertoire that exists only in paper or low-resolution scanned form. Public libraries, conservatory collections, and private fake-book editions contain hundreds of thousands of pages of notation that cannot be searched, transposed, played back, or analysed without manual transcription into notation software. Manual transcription is slow, expensive—typically requiring fifteen to sixty minutes per page depending on complexity—and demands specialist training that is unavailable at scale.

The practical consequences are significant. A music educator who wishes to assign a jazz standard in a different key must either own commercial software with the piece already encoded, transcribe it manually, or distribute a photocopy that students cannot edit. A musicologist studying harmonic patterns across a large corpus cannot apply statistical methods to scanned PDFs without first converting them to machine-readable format. A musician practising from a worn photocopy of *The Real Book* cannot feed the page to a practice app that transposes the melody to a B♭ instrument.

OCR technology solved an equivalent problem for text decades ago, making printed books, newspapers, and documents searchable and editable at almost zero marginal cost. OMR is the musical analogue, but it remains far less mature: the visual language of music notation is substantially more complex than alphabetic text, the available annotated data is orders of magnitude smaller, and the community of researchers and tools is correspondingly narrower.

1.1.2 Jazz Lead Sheets and *The Real Book*

Among the many formats of notated music, *jazz lead sheets* occupy a special position. A lead sheet is a compact score format that encodes only the most essential musical information: a single-voice melody line written on a staff, chord symbols printed above it indicating the harmonic context, and—occasionally—lyrics beneath the staff. This format is the lingua franca of jazz practice: it allows musicians to memorise and improvise over a tune from a minimal, portable representation without the burden of a full orchestral score.

The Real Book is the most widely circulated collection of jazz lead sheets. Originally assembled as an unauthorised compilation by Berklee College of Music students in the 1970s, it circulated as a hand-copied samizdat for years before receiving official publication. It contains several hundred of the most important jazz standards—tunes by Miles Davis, Thelonious Monk, Bill Evans, John Coltrane, and dozens of other canonical figures—and is found on the music stands of jazz musicians at virtually every level of practice, worldwide.

Despite its cultural centrality, the content of *The Real Book* remains almost entirely locked in paper or scanned-image form. The officially published editions exist as PDFs, but those PDFs are image-based: the musical content is not encoded as symbolic data, and standard music notation software cannot interpret them directly. Digitising *The Real Book*—and the broader class of jazz lead sheets it represents—would unlock a range of practical applications: automatic transposition for B♭, E♭, and F instruments; instant MIDI playback for practice and accompaniment; corpus-level harmonic analysis for musicologists; improved search and cataloguing in digital music libraries; and accessible preparation of educational materials for jazz educators.

1.1.3 Why Existing OMR Tools Fall Short

General-purpose OMR systems have advanced significantly in recent years [1], but they are designed and evaluated primarily on classical full-score notation: orchestral scores, piano scores, and choral settings printed in standard engraving styles (Finale, Sibelius, Dorico, or LilyPond's default font). Lead sheets diverge from this baseline in several ways that cause measurable degradation in existing tools.

Engraving style. Many editions of *The Real Book* use fonts and layouts that deliberately imitate handwritten notation. The LilyJAZZ font used by LilyPond—and the original hand-copy style of the Berklee editions—produce noteheads, stems, and clefs that differ visually from the upright, regularised glyphs of classical engraving software. Models trained on PrIMuS, DeepScoresV2 [2], or similar classical-style datasets do not transfer naturally to this visual register.

Chord symbols. Jazz chord symbols—comprising a root letter, quality abbreviation, extensions, alterations, and an optional slash bass note—appear above the staff as free-form text and are a core part of the lead-sheet semantics. Standard OMR pipelines are designed to transcribe pitch and rhythm from the staff; they either ignore chord symbols entirely or treat them as noise. Any OMR system targeting lead sheets must include a complementary chord recognition module that is coordinated with the melody transcription.

Absence of labelled data. To the best of the author's knowledge, no publicly available, labelled dataset of Real Book pages with ground-truth symbolic transcriptions exists as of 2026. Supervised training of modern deep learning models without such a corpus requires synthetic data generation and domain adaptation techniques whose quality directly governs the performance of the resulting system.

These three gaps—engraving style, chord semantics, and missing training data—define the specific challenges this thesis addresses and distinguish it from prior OMR work on classical notation.

1.2 Musical Notation Fundamentals

This section provides a self-contained introduction to the elements of Western staff notation relevant to this thesis. It is intended for readers with a computer science background who may not have formal training in music theory. Familiarity with this material is necessary to understand the vocabulary design, the data pipeline, and the error analysis presented in later chapters.

1.2.1 The Staff and the Treble Clef

Western music is written on a *staff*: a set of five equally spaced horizontal lines and the four spaces between them. Pitches are represented by the vertical position of a notehead on this grid: each line and space corresponds to a specific pitch, and moving a notehead one position upward (one line to the adjacent space, or one space to the adjacent line) raises the pitch by one diatonic step.

The absolute pitch assigned to each staff position depends on the *clef* symbol placed at the left edge of the staff. The most common clef in Western notation is the *treble clef* (also called the *G clef*), which curls around the second line from the bottom and designates that line as G4—the G above middle C. With this anchor established, the remaining lines read, from bottom to top, E4, G4, B4, D5, F5; the spaces between them read F4, A4, C5, E5.

When a pitch falls above or below the range of the five-line staff, short horizontal *ledger lines* extend the staff as needed. Middle C (C4), for example, sits on the first ledger line below the treble staff; a high A (A5) sits on the first ledger line above it. Jazz lead sheets use the treble clef exclusively, and this thesis targets the treble clef only (see Section 1.6).

Figure 1.1 illustrates the ascending C major scale from middle C (C4) to C5 on the treble staff. The leftmost note sits on a ledger line below the staff; successive notes step upward through the five-line grid, making the pitch–position correspondence explicit.

Figure 1.1. The C major scale (C4–C5) on the treble staff, rendered in LilyJAZZ notation. Middle C (C4) sits on the first ledger line below the staff; the remaining notes step upward through the lines and spaces, mapping each vertical position to a distinct pitch.

1.2.2 Pitch, Accidentals, and Key Signatures

Western tonal music divides the octave into twelve chromatic semitones. The seven diatonic pitch classes—A, B, C, D, E, F, G—correspond to the white keys of a piano keyboard and to the named staff positions described above. The five remaining semitones are represented using *accidentals*: a *sharp* (♯) raises a note by one semitone; a *flat* (♭) lowers it by one semitone; a *natural* (♮) cancels a preceding sharp or flat. Accidentals appear immediately to the left of the notehead they modify and remain in effect for the remainder of the bar in which they appear.

A *key signature*—placed at the beginning of each staff, immediately after the clef—specifies the tonal context of the piece by indicating which pitch classes are to be systematically sharpened or flattened throughout. A key signature of two sharps, for example, signals the key of D major and means that every F and every C should be played as F♯ and C♯ unless explicitly cancelled by a natural sign. Key signatures range from seven flats (C♭ major) to seven sharps (C♯ major), passing through C major, which carries no accidentals.

1.2.3 Note Duration and Rhythm

The *duration* of a note is encoded by a combination of three visual features:

Notehead shape. A hollow (open) notehead indicates a half note or whole note; a filled (solid) notehead indicates a quarter note or any shorter value. A whole note has no stem; all shorter notes have a vertical stem attached to the notehead.

Flags and beams. Each flag attached to a stem halves the duration of the note relative to a plain stemmed notehead: a single flag denotes an eighth note; two flags a sixteenth note; three flags a thirty-second note. When two or more short notes appear consecutively, their flags may be replaced by a horizontal *beam* connecting their stems, which clarifies the rhythmic grouping within the bar without changing the individual durations.

Augmentation dots. A dot placed immediately to the right of a notehead extends the note's duration by half its original value. A dotted quarter note therefore lasts for one and a half beats in common time; a dotted eighth note lasts for three sixteenth-note durations. Multiple dots are possible (each adding half the preceding augmentation), though double dots rarely appear in jazz lead sheets.

Rests indicate periods of silence and follow the same duration hierarchy as notes, with distinct graphical symbols for each value. A *whole-bar rest*—a filled rectangle hanging below a staff line—indicates that an entire measure is silent, regardless of the time signature.

Figure 1.2 shows the seven standard note values side by side, together with dotted variants. The progressive reduction in note value is visually encoded by the head shape (open vs. filled), stem flags, and augmentation dot.

Figure 1.2. Note duration values rendered in LilyJAZZ notation, from left to right: whole, half, quarter, eighth, 16th, 32nd, dotted quarter, and dotted eighth. The open notehead with no stem encodes the longest value (whole note); each flag added to the stem halves the duration; an augmentation dot extends it by half.

1.2.4 Time Signatures and Measures

The staff is divided by vertical *barlines* into segments called *measures* (or *bars*). Each measure contains a fixed total duration determined by the *time signature*, which appears at the beginning of a piece—and again whenever it changes—as a pair of stacked numbers. The upper number specifies how many beats are in each measure; the lower number specifies which note value counts as one beat (4 for a quarter note, 8 for an eighth note, 2 for a half note).

The time signature 4/4 ("common time") therefore assigns four quarter-note beats to every measure, while 3/4 assigns three, and 6/8 organises six eighth notes into two compound beats of three. The jazz repertoire is overwhelmingly written in 4/4, though 3/4 (waltz), 2/2 (cut time), 6/8, and 5/4 all appear in the Real Book.

1.2.5 Chord Symbols

Jazz notation supplements the melodic staff with *chord symbols* printed above the staff at positions corresponding, approximately, to the beat on which each harmonic change takes effect. These symbols specify the harmonic accompaniment and guide the rhythm section and improvising soloists.

A chord symbol encodes: (1) a *root pitch class* (A through G, with an optional flat or sharp); (2) an optional *quality abbreviation* specifying the chord type (e.g. `maj7` for major seventh, `m7` for minor seventh, `7` for dominant seventh, `m7b5` for half-diminished, `dim7` for fully diminished); (3) optional *extensions and alterations* modifying specific chord tones (e.g. `#9`, `b5`, `13`); and (4) an optional *slash bass note* indicating a non-root bass (e.g. `C/E` = C major chord over an E in the bass). Representative examples from *The Real Book* include: `Cmaj7`, `F#m7b5`, `D7#9`, `Bb13`, `Am7/D`.

Chord symbols are typeset as free-form text—not as graphical notation symbols—and their recognition requires optical character recognition (OCR) rather than the staff-pattern recognition applied to the melody line. This distinction motivates the two-stream architecture described in Chapter 4.

1.2.6 The Lead-Sheet Format

A complete jazz lead sheet consists of: (1) one or more systems of single-voice treble-clef staves carrying the melody, with clef, key signature, and time signature at the start of each system; (2) chord symbols above each staff, roughly aligned with the beat at which each harmonic change takes effect; and (3) occasionally, structural markers such as repeat signs, section labels (*A*, *B*, *Bridge*), and tempo or style indications—though these are absent from many informal editions, including large parts of *The Real Book*.

Figure 1.3 illustrates a clean synthetic rendering of a lead-sheet excerpt in the LilyJAZZ style used throughout this project, and Figure 1.4 shows an actual page from *The Real Book* with its key notation elements identified.

Figure 1.3. A four-bar jazz lead-sheet excerpt in C major (4/4), rendered in LilyJAZZ style—the same engraving used to generate the synthetic training dataset. Chord symbols above the staff (`Cmaj7`, `Am7`, `Dm7`, `G7`) encode the I–vi–ii–V harmonic progression; the melody line encodes pitch through vertical position and duration through notehead shape and stem.

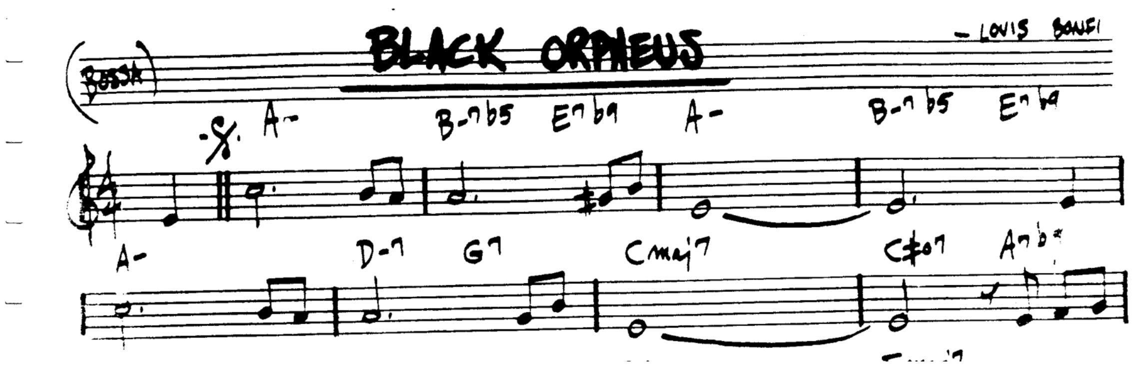

Figure 1.4. Excerpt from *The Real Book* (“Black Orpheus”, L. Bonfá), showing the hand-engraved notation the OMR system must transcribe: a treble staff with key and time signatures, melody notes of varying durations, and chord symbols written above the staff.

1.3 Problem Formulation

The goal of this thesis is to build a system that, given a PDF or image of a jazz lead sheet, automatically produces a structured, machine-readable transcription of its contents: the sequence of notes, rests, and rhythmic markers making up the melody, and the chord symbols specifying the harmonic progression. Despite decades of OMR research, reliable automated transcription of real-world scores remains an open problem [1]. The challenge has several compounding sources of difficulty.

Notation complexity. Unlike alphabetic text, where symbols are largely independent and arranged in a single linear sequence, musical notation encodes information jointly across two spatial dimensions. A note's pitch is determined by its vertical position on the staff *relative to the clef*, not by the symbol shape alone; its duration is determined by a combination of head shape, stem presence, flag count, and dot presence. Beams create joint dependencies across groups of notes; ties extend a single pitch across a barline. Any recognition system must capture these two-dimensional, context-dependent symbol interactions without relying on explicit per-symbol segmentation at inference time.

Domain gap. The most widely used annotated OMR datasets—including PrIMuS [3] and DeepScoresV2 [2]—consist of cleanly computer-engraved images rendered at high resolution with standard engraving fonts. Real-world documents, by contrast, are photographed under imperfect lighting conditions, and suffer from scanner noise, perspective distortion, ink bleed, paper yellowing, and JPEG compression artefacts. A model trained exclusively on clean synthetic data generalises poorly to real scans: the feature distribution shifts substantially when crisp black symbols on a white background are replaced by blurred, uneven, noisy images with variable contrast. Bridging this *domain gap* without real annotated training data requires a scan-simulation augmentation pipeline that teaches the model to be invariant to scan-related distortions.

Domain specificity. General-purpose OMR tools are designed and evaluated on classical full-score notation. Jazz lead sheets use a different engraving style—frequently imitating hand-lettered notation with rounded noteheads, loosely spaced beams, and a distinctive chord-symbol typography—for which no publicly labelled corpus currently exists. Training a model for this specific domain therefore requires constructing a new synthetic dataset from scratch, using a rendering pipeline tuned to the jazz style.

Dual-stream recognition. The lead-sheet format requires simultaneously processing two distinct visual streams from the same image region: the melodic staff content, which calls for pattern-based sequence recognition, and the chord symbols above the staff, which call for text OCR. These two streams require different feature representations and must ultimately be integrated into a coherent structured output. A complete lead-sheet OMR system must solve both sub-problems and align their results.

This thesis addresses all four difficulties jointly: by constructing a synthetic training corpus in the jazz engraving style, applying a calibrated scan-simulation augmentation pipeline, training a CRNN-CTC model for melody recognition, and building a separate OCR pathway for chord symbols that is fused into a unified output by the inference pipeline.

1.4 Objectives

The **general objective** of this project is to develop, evaluate, and document a proof-of-concept end-to-end OMR system for monophonic jazz lead sheets, and to demonstrate that a CRNN-CTC architecture trained on synthetic data can bridge the domain gap between clean rendered images and real scanned pages.

The **specific objectives** are:

1. Survey the state of the art in OMR, covering traditional modular pipelines, CRNN-CTC sequence models, and transformer-based end-to-end approaches, and justify the architectural

choices made in this project.

2. Design and implement a synthetic training dataset by rendering excerpts from the PrIMuS corpus [3] in a jazz engraving style using LilyPond and the LilyJAZZ font, paired with ground-truth annotations in the LMX token format [4].

3. Design and implement a scan-simulation augmentation pipeline calibrated against real Real Book scan characteristics, to close the domain gap at training time without requiring manually labelled real-world data.

4. Design, implement, and train a CRNN-CTC model for monophonic staff-line recognition, and evaluate it rigorously using the *Symbol Error Rate* (SER) [5] metric.

5. Perform quantitative error analysis to characterise systematic failure modes and guide iterative data and model refinements.

6. Build a complete inference pipeline—encompassing PDF input handling, staff detection, melody recognition, grammar correction, and chord symbol extraction—that returns structured JSON output suitable for downstream musical applications.

7. Quantify the domain gap between synthetic and real-scan performance, and reduce it via a fine-tuning stage on a small set of manually annotated Real Book pages.

1.5 Proposed Approach

The core design choice of this thesis is a *CRNN-CTC* architecture: a Convolutional Recurrent Neural Network trained with the Connectionist Temporal Classification (CTC) loss. This model class is well-matched to the left-to-right, variable-length token sequences that arise from reading a single staff line from left to right: the convolutional backbone extracts local visual features along the width of the staff image; two bidirectional LSTM layers model long-range sequential dependencies; and CTC decoding maps the resulting frame-level distributions to a token sequence without requiring explicit segmentation of the image into individual symbols.

An alternative—transformer-based end-to-end models such as the Sheet Music Transformer [6] or the OLIMPIC system for pianoform music [7]—offers higher modelling capacity but demands substantially larger labelled datasets and compute budgets than are available within the scope of a Bachelor's thesis. Traditional heuristic pipelines [8] avoid data requirements but accumulate segmentation errors that propagate irreversibly through the recognition chain. The CRNN-CTC architecture represents the best trade-off for this project's constraints; this judgement is substantiated by the comparative analysis in Chapter 3.

The system is organised around a five-stage inference pipeline: (1) *Preprocessing* binarises and deskews the input image; (2) *Staff detection* locates individual staff systems using morphological operations; (3) *Melody recognition* applies the CRNN-CTC model to each isolated staff image and produces a raw token sequence in LMX format; (4) *Grammar correction* validates the sequence against the LMX grammar and repairs structurally ill-formed outputs; (5) *Chord OCR* extracts chord symbols from the strip above each staff and post-processes them. The outputs of stages 4 and 5 are merged into the structured JSON format described in Chapter 4.

The training strategy also deserves mention. In the absence of labelled real-world data, the model is first trained entirely on synthetic LilyJAZZ-rendered images augmented with scan simulation, then fine-tuned on a small annotated set of real Real Book pages. This two-phase approach is a standard domain adaptation technique and is the practical solution to the data scarcity problem described in Section 1.3.

1.6 Scope

The project is bounded by the following scope definition.

In scope:

- Monophonic (single-voice) melody lines in treble clef, as found in jazz lead sheets.
- Synthetic training data construction via LilyPond rendering of PrIMuS excerpts in the LilyJAZZ engraving style.
- Scan-simulation augmentation modelling noise, blur, perspective distortion, ink bleed, and photometric variation.
- Staff region detection as a preprocessing module.
- Design, training, and evaluation of a CRNN-CTC model for single-staff melody recognition.
- Chord symbol extraction via an OCR-based module.
- Quantitative evaluation using SER and qualitative error analysis.
- Domain-gap analysis and fine-tuning on a small set of annotated Real Book pages.
- A FastAPI web service exposing the full pipeline as a REST endpoint.

Out of scope:

- Polyphonic transcription or simultaneous multi-voice recognition.
- Full-page layout analysis and reading-order recovery across multiple stave systems beyond the per-staff processing implemented.
- Handwritten scores (as opposed to printed or typeset notation).
- Non-Western notation systems or extended contemporary notation.
- Real-time or mobile-device deployment.
- Training large-scale transformer models from scratch.
- Multi-page PDF processing (currently limited to page 0 of any multi-page document).
- Tuplets, repeat barlines, voltas, and navigation marks (D.C., D.S., segno, coda), which do not appear frequently enough in the in-scope repertoire to justify the vocabulary and grammar extensions they would require.

1.7 Document Structure

The remainder of this document is organised as follows.

Chapter 2 surveys the state of the art in OMR, covering the progression from heuristic modular pipelines through CRNN-CTC models to modern transformer-based end-to-end architectures, and reviews the available datasets and symbolic representations relevant to this work.

Chapter 3 covers project management: the justification of the chosen architecture through comparative analysis, the work-breakdown structure and timeline, the budget and economic analysis, and the project's sustainability and ethical impact assessment.

Chapter 4 describes the technical methodology in full detail: the data generation and augmentation pipeline, the LMX vocabulary and token grammar, the CRNN-CTC architecture and training procedure, and the complete inference pipeline.

Chapter 5 presents experimental results: training curves, SER on the synthetic test set, ablation studies, qualitative error analysis by failure category, and domain-gap measurements with and without fine-tuning on real Real Book pages.

Chapter 6 revisits sustainability and ethical considerations in light of the completed work.

Finally, Chapter 7 draws conclusions, reflects on the extent to which the stated objectives were achieved, and outlines directions for future work.

Chapter 2

State of the Art

2.1 Historical Approaches

[TODO: ***Traditional heuristic/rule-based pipelines; staff detection/removal; why OpenCV morphology was insufficient.*****]**

2.2 Deep Learning Paradigm: CRNN and CTC

[TODO: ***CRNN as visual encoder + sequence model; CTC alignment; why it fits left-to-right monophonic lines.*****]**

2.3 Modern End-to-End Architectures

[TODO: ***Transformer-based OMR (Sheet Music Transformer, LEGATO, etc.) and trade-offs.*****]**

2.4 Output Representations

[TODO: ***LMX, **kern, agnostic vs. semantic; why LMX linearisation is used here.*****]**

Chapter 3

Project Management

3.1 Justification of Alternatives

[TODO: ***CRNN–CTC vs. classic CV vs. large transformers; constraints and rationale.*****]**

3.2 Project Planning

[TODO: ***Gantt, milestones, deviations/risks; link to GEP deliverables.*****]**

3.3 Economic Management

[TODO: ***Budget: hardware, hours costed as junior engineer, licences (if any), contingency.*****]**

Chapter 4

Methodology and Architecture

4.1 The Data Pipeline

4.1.1 Synthetic Generation with LilyPond and LilyJAZZ

[**TODO:** ***Rendering pipeline; notation constraints; dataset size targets.***]

4.1.2 Data Cleaning

[**TODO:** ***Filtering degenerate samples; criteria; impact (e.g., 26k removed).***]

4.1.3 Noise Engine

[**TODO:** ***Perspective warp, blur, scan-like artefacts; calibration vs. Real Book scans.***]

4.2 Neural Network Architecture

4.2.1 ResNet-18 Backbone

[**TODO:** ***Custom ResNet-18; feature map shapes; why chosen.***]

4.2.2 Asymmetric Strides and Temporal Resolution

[**TODO:** ***Height-to-1 squashing while preserving width for CTC time steps; math/intuition.***]

4.3 Training Loop

[**TODO:** ***PyTorch AMP, OneCycleLR, early stopping, checkpoints, config choices.***]

4.4 Bridging the Domain Gap (Fine-Tuning)

[**TODO:** ***Small Real Book dataset; freezing early layers; transfer learning strategy.***]

Chapter 5

Results and Evaluation

5.1 Synthetic Baseline Results

[**TODO:** *SER improvements (e.g., 7.4% → 1.17%); tables/plots; ablations.*]

5.2 The Domain Gap (Zero-Shot Testing)

[**TODO:** *Performance on real handwritten scan pre-fine-tuning; qualitative errors.*]

5.3 Out-Of-Vocabulary (OOV) Solutions

[**TODO:** *Pipeline updates; normalisation strategy (e.g., repeat signs → barlines).*]

5.4 Fine-Tuning Results

[**TODO:** *Post fine-tuning evaluation; Autumn Leaves read without text hallucinations.*]

Chapter 6

Sustainability and Ethical Implications

6.1 Environmental Sustainability

[TODO: ***Training energy, GPU hours, carbon footprint; mitigations: AMP, early stopping.*****]**

6.2 Economic Sustainability

[TODO: ***Cost/benefit vs. commercial tools and manual transcription.*****]**

6.3 Social Sustainability and Ethics

[TODO: ***Open-sourcing, accessibility, education; data and licensing considerations.*****]**

Chapter 7

Conclusions and Future Work

7.1 Conclusions

[TODO: ***Key achievements; framing as Seq2Seq; synthetic SER; domain-gap fine-tuning.*****]**

7.2 Future Lines of Research

[TODO: ***Full-page transcription (layout analysis / YOLO), polyphony, broader vocab, transformers.*****]**

References

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Appendix A

Additional Tables and Hyperparameters

[TODO: ***Hyperparameter sweeps, training configs, etc.*****]**

Appendix B

Qualitative Predictions

[TODO: ***Extra figures and predictions that would clutter the main text.*****]**

Appendix C

Code Snippets

[TODO: ***Only the most relevant snippets; keep main text clean.*****]**